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Inventor(s)	Silvestri; Claudio et al.

Dual chamber blood reservoir

Abstract

A blood reservoir may be used in combination with other elements such as a heart lung machine (HLM), oxygenator, heat exchanger, arterial filter and the like to form an extracorporeal blood circuit that may be employed in a procedure such as a bypass procedure. The blood reservoir may be configured to receive, filter and store blood from a number of sources including vent blood (from within the heart), venous blood (from a major vein), purge blood (from a sampling line) and cardiectomy or suction blood (from the surgical field).

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3551072	12/1969	Zemmerly	N/A	N/A
3588589	12/1970	Vonk	N/A	N/A
3588859	12/1970	Petree	N/A	N/A
3851181	12/1973	Heule	N/A	N/A
3927980	12/1974	Leonard	N/A	N/A
4006745	12/1976	Sorenson et al.	N/A	N/A
4170765	12/1978	Austin et al.	N/A	N/A
4177649	12/1978	Venema	N/A	N/A
4193004	12/1979	Lobdell et al.	N/A	N/A
4275726	12/1980	Schael	N/A	N/A
4309871	12/1981	Venema	N/A	N/A
4374088	12/1982	Stenberg et al.	N/A	N/A
4464164	12/1983	Troutner et al.	N/A	N/A
4466804	12/1983	Hino	N/A	N/A
4490331	12/1983	Steg, Jr.	N/A	N/A
4518318	12/1984	Jensen et al.	N/A	N/A
4530696	12/1984	Bisera et al.	N/A	N/A
4599093	12/1985	Steg, Jr.	N/A	N/A
4602344	12/1985	Ferretti et al.	N/A	N/A
4642089	12/1986	Zupkas et al.	N/A	N/A
4664682	12/1986	Monzen	N/A	N/A
4678404	12/1986	Lorett et al.	N/A	N/A
4701101	12/1986	Sapoff	N/A	N/A
4705497	12/1986	Shitaokoshi et al.	N/A	N/A
4782451	12/1987	Mazzarella et al.	N/A	N/A
4828543	12/1988	Weiss et al.	N/A	N/A

4846800	12/1988	Ouriel et al.	N/A	N/A
4876066	12/1988	Bringham et al.	N/A	N/A
4955874	12/1989	Farrar et al.	N/A	N/A
4984462	12/1990	Hass, Jr. et al.	N/A	N/A
4991433	12/1990	Warnaka et al.	N/A	N/A
5039430	12/1990	Corey, Jr.	N/A	N/A
5039482	12/1990	Panzani et al.	N/A	N/A
5043707	12/1990	Heinze	N/A	N/A
5049146	12/1990	Bringham et al.	N/A	N/A
5055198	12/1990	Shettigar	N/A	N/A
5060512	12/1990	Kanashige et al.	N/A	N/A
5061236	12/1990	Sutherland et al.	N/A	N/A
5078677	12/1991	Gentelia et al.	N/A	N/A
5110549	12/1991	Gordon	N/A	N/A
5112480	12/1991	Hukasawa	N/A	N/A
5120303	12/1991	Hombrouckx	N/A	N/A
5135485	12/1991	Cohen et al.	N/A	N/A
5147187	12/1991	Ito et al.	N/A	N/A
5149318	12/1991	Lindsay	N/A	N/A
5158533	12/1991	Strauss	210/489	A61M 1/3627
5178603	12/1992	Prince	N/A	N/A
5186431	12/1992	Tamari	N/A	N/A
5215519	12/1992	Shettigar	N/A	N/A
5226265	12/1992	Kelly et al.	N/A	N/A
5240380	12/1992	Mabe	N/A	N/A
5266265	12/1992	Raible	N/A	N/A
5270005	12/1992	Raible	N/A	N/A
5282783	12/1993	Lindsay	N/A	N/A
5303585	12/1993	Lichte	N/A	N/A
5304164	12/1993	Lindsay	N/A	N/A
5318510	12/1993	Cathcart	N/A	N/A
5399074	12/1994	Nose et al.	N/A	N/A
5403273	12/1994	Lindsay	N/A	N/A
5411705	12/1994	Thor et al.	N/A	N/A
5458566	12/1994	Herrig et al.	N/A	N/A
5458567	12/1994	Cathcart	N/A	N/A
5458579	12/1994	Chodorow et al.	N/A	N/A
5563490	12/1995	Kawaguchi et al.	N/A	N/A
5563584	12/1995	Rader et al.	N/A	N/A
5586085	12/1995	Lichte	N/A	N/A
5591399	12/1996	Goldman et al.	N/A	N/A
5604315	12/1996	Briefer et al.	N/A	N/A
5619993	12/1996	Lee	N/A	N/A
5667485	12/1996	Lindsay	N/A	N/A
5725357	12/1997	Nakazeki et al.	N/A	N/A
5756940	12/1997	Van Driel et al.	N/A	N/A
5770073	12/1997	Bach et al.	N/A	N/A
5775879	12/1997	Durando	N/A	N/A
5800721	12/1997	McBride	N/A	N/A

5823045	12/1997	Van Driel et al.	N/A	N/A
5826576	12/1997	West	N/A	N/A
5849186	12/1997	Raneri	210/489	A61M 1/3623
5928180	12/1998	Krivitski et al.	N/A	N/A
5955672	12/1998	Van Driel et al.	N/A	N/A
6017493	12/1999	Cambron et al.	N/A	N/A
6048363	12/1999	Nagyszalanczy et al.	N/A	N/A
6123519	12/1999	Kato et al.	N/A	N/A
6146411	12/1999	Noda et al.	N/A	N/A
6164325	12/1999	Braun	N/A	N/A
6287270	12/2000	Fini	N/A	N/A
6337049	12/2001	Tamari	N/A	N/A
6345214	12/2001	Dulphy-Vigor et al.	N/A	N/A
6475176	12/2001	Fini	N/A	N/A
6542848	12/2002	Neeser et al.	N/A	N/A
6562012	12/2002	Brown et al.	N/A	N/A
6564627	12/2002	Sabini et al.	N/A	N/A
6592340	12/2002	Horo et al.	N/A	N/A
6631639	12/2002	Dam et al.	N/A	N/A
6652495	12/2002	Walker	N/A	N/A
6694570	12/2003	Chen	N/A	N/A
6770048	12/2003	Fini	N/A	N/A
6931926	12/2004	Van Ee	N/A	N/A
7072769	12/2005	Fletcher-Haynes et al.	N/A	N/A
7147614	12/2005	Fini	N/A	N/A
7591812	12/2008	Tamari	N/A	N/A
7694570	12/2009	Dam et al.	N/A	N/A
7982612	12/2010	Braun	N/A	N/A
8105265	12/2011	Demers et al.	N/A	N/A
8394321	12/2012	Franzoni et al.	N/A	N/A
8409124	12/2012	Steffens et al.	N/A	N/A
8500673	12/2012	Zanotti et al.	N/A	N/A
8506513	12/2012	Rossi et al.	N/A	N/A
8734376	12/2013	Simpson et al.	N/A	N/A
9011769	12/2014	Silvestri	604/6.14	A61M 1/3638
10213541	12/2018	Silvestri	N/A	A61M 1/3666
10458833	12/2018	Rossi	N/A	N/A
11389580	12/2021	Silvestri	N/A	A61M 1/3666
2001/0013822	12/2000	Nazarian et al.	N/A	N/A
2001/0050256	12/2000	Krivitski	N/A	N/A
2002/0032399	12/2001	Fini	N/A	N/A
2002/0033181	12/2001	Groth et al.	N/A	N/A
2002/0038392	12/2001	De La Huerga	N/A	N/A
2002/0085952	12/2001	Ellingboe et al.	N/A	N/A

2002/0094300	12/2001	Yokoyama	604/406	A61M 1/3627
2002/0128582	12/2001	Farrell et al.	N/A	N/A
2002/0133066	12/2001	Miller et al.	N/A	N/A
2003/0033871	12/2002	Carroll et al.	N/A	N/A
2003/0035730	12/2002	Schob	N/A	N/A
2003/0045772	12/2002	Reich et al.	N/A	N/A
2003/0139643	12/2002	Smith et al.	N/A	N/A
2003/0144646	12/2002	Se et al.	N/A	N/A
2003/0175151	12/2002	Ghelli et al.	N/A	N/A
2004/0047737	12/2003	Nose et al.	N/A	N/A
2004/0064292	12/2003	Beck et al.	N/A	N/A
2004/0152944	12/2003	Medvedev et al.	N/A	N/A
2005/0025630	12/2004	Ayre et al.	N/A	N/A
2005/0119600	12/2004	Lucke et al.	N/A	N/A
2005/0230313	12/2004	O'Mahony et al.	N/A	N/A
2006/0015056	12/2005	Ellingboe et al.	N/A	N/A
2006/0089695	12/2005	Bolea et al.	N/A	N/A
2006/0092360	12/2005	Hong	N/A	N/A
2006/0150596	12/2005	Takahashi et al.	N/A	N/A
2006/0167400	12/2005	Ellingboe et al.	N/A	N/A
2006/0226087	12/2005	Robinson et al.	N/A	N/A
2006/0260392	12/2005	Hedrick	N/A	N/A
2006/0277269	12/2005	Dent et al.	N/A	N/A
2007/0017518	12/2006	Farrugia et al.	N/A	N/A
2007/0110612	12/2006	Ito	N/A	N/A
2007/0142923	12/2006	Ayre et al.	N/A	N/A
2007/0194981	12/2006	Hagg et al.	N/A	N/A
2007/0209662	12/2006	Bowen et al.	N/A	N/A
2008/0027368	12/2007	Kollar et al.	N/A	N/A
2008/0078382	12/2007	LeMahieu et al.	N/A	N/A
2008/0171960	12/2007	Brieske et al.	N/A	N/A
2008/0245530	12/2007	Kuzmichev	N/A	N/A
2008/0275377	12/2007	Paolini et al.	N/A	N/A
2009/0012443	12/2008	Ghelli et al.	N/A	N/A
2009/0099498	12/2008	Demers et al.	N/A	N/A
2009/0149950	12/2008	Wampler	N/A	N/A
2010/0042038	12/2009	Urdahl et al.	N/A	N/A
2010/0140182	12/2009	Chapman et al.	N/A	N/A
2010/0275953	12/2009	Orue et al.	N/A	N/A
2010/0280430	12/2009	Caleffi et al.	N/A	N/A
2011/0098625	12/2010	Masala et al.	N/A	N/A
2011/0257576	12/2010	Simpson et al.	N/A	N/A
2011/0257578	12/2010	Zanotti et al.	N/A	N/A
2011/0257579	12/2010	Rossi et al.	N/A	N/A
2012/0067133	12/2011	Waldrop et al.	N/A	N/A
2012/0130299	12/2011	Knott et al.	N/A	N/A
2012/0226446	12/2011	Nelson et al.	N/A	N/A
2013/0017119	12/2012	Silvestri et al.	N/A	N/A
2013/0303965	12/2012	Rossi et al.	N/A	N/A

2013/0331758	12/2012	Meibaum et al.	N/A	N/A
2014/0278156	12/2013	Thompson et al.	N/A	N/A
2015/0100253	12/2014	Austerlitz et al.	N/A	N/A
2015/0196703	12/2014	Silvestri et al.	N/A	N/A
2015/0367120	12/2014	Kusters et al.	N/A	N/A
2017/0089746	12/2016	Rossi	N/A	N/A
2019/0070353	12/2018	Knott et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
86103696	12/1986	CN	N/A
1147964	12/1996	CN	N/A
1197677	12/1997	CN	N/A
1458851	12/2002	CN	N/A
2455229	12/1975	DE	N/A
2754894	12/1978	DE	N/A
3935502	12/1990	DE	N/A
19840399	12/1998	DE	N/A
102004040441	12/2005	DE	N/A
102005001779	12/2005	DE	N/A
102005029682	12/2005	DE	N/A
102007026010	12/2009	DE	N/A
0371173	12/1989	EP	N/A
0587251	12/1993	EP	N/A
0472480	12/1994	EP	N/A
0820775	12/1997	EP	N/A
0952433	12/1998	EP	N/A
1053760	12/1999	EP	N/A
1070509	12/2000	EP	N/A
0690730	12/2001	EP	N/A
1210956	12/2001	EP	N/A
1003575	12/2003	EP	N/A
0766974	12/2005	EP	N/A
2754458	12/2013	EP	N/A
2435106	12/2013	EP	N/A
2842584	12/2014	EP	N/A
2811752	12/2001	FR	N/A
2009862	12/1978	GB	N/A
2109934	12/1982	GB	N/A
S5623960	12/1980	JP	N/A
S57500411	12/1981	JP	N/A
S62258671	12/1986	JP	N/A
H0391352	12/1990	JP	N/A
H0819602	12/1995	JP	N/A
H08506982	12/1995	JP	N/A
H11506701	12/1998	JP	N/A
2944749	12/1998	JP	N/A
20000000299	12/1999	JP	N/A
2001503665	12/2000	JP	N/A
2001165878	12/2000	JP	N/A

2001204815	12/2000	JP	N/A
2001514939	12/2000	JP	N/A
2001523339	12/2000	JP	N/A
200233648	12/2001	JP	N/A
2003052717	12/2002	JP	N/A
2003126246	12/2002	JP	N/A
2005066013	12/2004	JP	N/A
2006025531	12/2005	JP	N/A
2006325750	12/2005	JP	N/A
2007130290	12/2006	JP	N/A
2008000597	12/2007	JP	N/A
2008194386	12/2007	JP	N/A
2008270595	12/2007	JP	N/A
2009240428	12/2008	JP	N/A
2009287593	12/2008	JP	N/A
2011076394	12/2010	JP	N/A
9965413	12/1998	NO	N/A
9421311	12/1993	WO	N/A
9624397	12/1995	WO	N/A
9733672	12/1996	WO	N/A
9820957	12/1997	WO	N/A
9848868	12/1997	WO	N/A
9908734	12/1998	WO	N/A
0015154	12/1999	WO	N/A
0044415	12/1999	WO	N/A
0147442	12/2000	WO	N/A
0176656	12/2000	WO	N/A
0239931	12/2001	WO	N/A
0239933	12/2001	WO	N/A
02095675	12/2001	WO	N/A
03026724	12/2002	WO	N/A
2006021295	12/2005	WO	N/A
2006057650	12/2005	WO	N/A
2006122282	12/2005	WO	N/A
2007018513	12/2006	WO	N/A
2008119993	12/2007	WO	N/A
2009144522	12/2008	WO	N/A
2010041604	12/2009	WO	N/A
2014041604	12/2013	WO	N/A

OTHER PUBLICATIONS

Catalog of Products, Terumo Europe Cardiovascular Systems, 142 pages, 2009. cited by applicant
Definition of “Cylinder”, downloaded from <http://dictionary.reference.com/browse/cylinder>, 3 pages, download on Apr. 28, 2014. cited by applicant

European Search Report and Search Opinion Received for EP Application No. 14164506.9, dated Sep. 19, 2014, 10 pages. cited by applicant

European Search Report issued in EP Application No. 11173655, completed Nov. 30, 2011, 9 pages. cited by applicant

Extended European Search Report issued in 14188440.3, dated Jan. 30, 2015, 7 pages. cited by applicant

Fischer, Betriebsmesstechnik, unveränderte Auflage, VEB Verlag Technik Berlin, 3 pages (machine translations: Business measuring technique, unchanged edition). 1986. cited by applicant

Henriksen et al; “Envisioning Patient Safety in the Year 2025: Eight Perspectives,” Advances in Patient Safety: New Directions and Alternative Approaches, Agency for Healthcare Research and Quality, vol. 1, Aug. 2008. cited by applicant

International Preliminary Report on Patentability issued in PCT/IB2014/061491, dated Dec. 1, 2016, 12 pages. cited by applicant

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/IB2011/051639, dated Nov. 1, 2012 10 pages. cited by applicant

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/IB2012/053497, dated Jan. 23, 2014, 10 pages. cited by applicant

International Preliminary Report on Patentability, Chapter 11, issued in PCT/EP2010/055522, (with translation), dated May 31, 2013 13 pages. cited by applicant

International Search Report and Written Opinion issued in PCT/EP2010/055522, (with translation) dated Aug. 6, 2010, 10 pages. cited by applicant

International Search Report and Written Opinion issued in PCT/IB2011/051639, dated Nov. 18, 2011, 15 pages. cited by applicant

International Search Report and Written Opinion issued in PCT/IB2014/061491, dated Mar. 6, 2015, 16 pages. cited by applicant

International Search Report and Written Opinion received for PCT Patent Application No. PCT/IB2012/053497, dated Nov. 15, 2012. 12 pages. cited by applicant

Klonoff, Designing an Artificial Pancreas System to be Compatible with Other Medical Devices, Journal of Diabetes Science and Technology, vol. 2, No. 5, pp. 741-745, Sep. 2008. cited by applicant

Terumo Europe Cardiovascular Systems, Innovative Products for the Treatment of Cardiovascular Disease, 2006 Terumo Europe, 105 pages, 2006. cited by applicant

Van Der Togt et al, “Electromagnetic Interference from Radio Frequency Identification Inducing Potentially Hazardous Incidents in Critical Care Medical Equipment,” JAMA, vol. 299, No. 24, 7 pages, Jun. 25, 2008. cited by applicant

Weber, “Talking Barcodes that Change our Lives,” BBC News, 3 pages, Published Apr. 28, 2004. (check date). cited by applicant

Wikipedia, “Fullstandmeussing” [online]. Retrieved from <https://de.wikipedia.org/w/index.php?title=F%C3%BCIstandmessung&oldid=69-998631>, last modified Jan. 30, 2010. English translation retrieved from https://en.wikipedia.org/wiki/Level_sensor, Oct. 18, 2016. cited by applicant

International Search and Written Opinion issued in PCT/IB2014/061491, dated Mar. 8, 2015, 16 pages. cited by applicant

International Preliminary Report on Patentability issued in PCT/EP2010/055444, dated Oct. 5, 2011, 10 pages (with English translation). cited by applicant

International Search Report and Written Opinion issued in PCT/EP2010/055444, dated Aug. 20, 2010, 10 pages (English Translation of SR). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 16/273,459 filed Feb. 12, 2019, which is a continuation of U.S. application Ser. No. 14/668,933 filed Mar. 25, 2015, now U.S. Pat. No. 10,213,541, which is a division of U.S. application Ser. No. 13/181,688, filed Jul. 13, 2011, now U.S. Pat. No. 9,011,769, which claims priority to European Patent Application 11173655.9, filed Jul. 12, 2011, all of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present invention relates generally to blood reservoirs for oxygenators used in blood perfusion systems.

BACKGROUND

(2) Blood perfusion involves encouraging blood through the vessels of the body. For such purposes, blood perfusion systems typically include the use of one or more pumps in an extracorporeal circuit that is interconnected with the vascular system of a patient. Many surgical procedures require or prefer temporary cessation of the heart to create a still operating field. Such procedures may thus rely upon a cardiopulmonary bypass (CPB) perfusion system that temporarily replaces the function of the heart and lungs. Examples of such procedures include the surgical correction of vascular stenosis, valvular disorders, and congenital heart defects. In perfusion systems used for cardiopulmonary bypass surgery, an extracorporeal blood circuit is established that includes at least one pump and an oxygenation device to replace the functions of the heart and lungs, respectively.

(3) More specifically, in cardiopulmonary bypass procedures, oxygen-poor blood (i.e., venous blood) is gravity-drained or vacuum-suctioned from a large vein entering the heart or another major vein in the body (e.g., femoral) and is transferred through a venous line in the extracorporeal circuit. The venous blood is pumped to an oxygenator that provides for oxygen transfer to the blood. Oxygen may be introduced into the blood by, for example, transfer across a membrane. Concurrently, carbon dioxide is removed across the membrane. The oxygenated blood is filtered and then returned through an arterial line to the aorta, femoral, or other artery.

(4) In many cases, an extracorporeal blood circuit includes a blood reservoir that can be used to collect, filter and de-aerate blood from a variety of different sources. For example, a blood reservoir may receive one or more of venous blood from a large vein, vent blood that is collected within the heart and cardiectomy or suction blood that is collected from outside the heart but within the surgical field.

SUMMARY

(5) The present invention relates to a blood reservoir that may be used in combination with other elements such as a heart lung machine (HLM), oxygenator, heat exchanger, arterial filter and the like to form an extracorporeal blood circuit. The blood reservoir, as will be described in greater detail herein, may be configured to receive, filter and store blood from a number of sources including vent blood (from within the heart), venous blood (from a major vein), purge blood (from a sampling line) and cardiectomy or suction blood (from within the surgical field). Example 1 is a dual chamber blood reservoir including an activated section and a non-activated section. The non-activated, or clean, section includes an elongate filter and a foamer that is disposed about an upper region of the elongate filter. A purgers funnel extends downwardly through the cylindrical foamer and includes a conical upper portion, a cylindrical lower portion and an intervening central (6) portion. A venous inlet tube extends downwardly through the cylindrical lower portion of the purgers funnel to a position that is proximate a bottom surface of the elongate filter. A vent inlet tube extends downwardly through an aperture formed within the central portion of the purgers funnel to a position that is proximate the bottom surface of the elongate filter.

(7) In Example 2, the dual chamber blood reservoir of Example 1 in which blood that exits the cylindrical lower portion of the purgers funnel is able to slide down the exterior surface of the venous inlet tube.

- (8) In Example 3, the dual chamber blood reservoir of Example 1 or 2 in which the central portion of the purgers funnel includes a first aperture that is configured to accommodate the vent inlet tube passing therethrough.
- (9) In Example 4, the dual chamber blood reservoir of any of Examples 1, 2 or 3, further including a second vent inlet tube that extends downwardly to a position that is proximate the bottom of the elongate filter.
- (10) In Example 5, the dual chamber blood reservoir of Example 4, wherein the central portion of the purgers funnel includes a second aperture that is configured to accommodate the second vent inlet tube, the first and second apertures being radially spaced apart about 180 degrees.
- (11) In Example 6, the dual chamber blood reservoir of any of Examples 1 to 5, further including a plurality of purge ports that are in fluid communication with the conical upper portion of the purgers funnel.
- (12) In Example 7, the dual chamber blood reservoir of any of Examples 1 to 6 in which the activated section includes a suction blood filter assembly including a cylindrical suction blood filter and a defoamer layer that is disposed about the cylindrical suction blood filter.
- (13) In Example 8, the dual chamber blood reservoir of any of Examples 1 to 7, further including a releasable barrier between the activated section and the non-activated section, the releasable barrier configured to be released to permit blood within the activated section to enter the non-activated section in a situation requiring additional blood.
- (14) In Example 9, the dual chamber blood reservoir of Example 8, further including a porous media disposed to dissipate velocity in blood flowing from the activated section to the non-activated section.
- (15) Example 10 is a dual chamber blood reservoir having a housing and a cover spanning the housing. A first vent port and a second vent port each extend through the cover. A venous port extends through the cover. A purgers port extends through the cover. The blood reservoir includes a purgers funnel that has an upper portion, a lower portion and an intervening central portion. The upper portion is in fluid communication with the purgers port. A first vent tube is in fluid communication with the first vent port and extends externally to the lower portion of the purgers funnel to a position near a lower surface of the housing. A second vent tube is in fluid communication with the second vent port and extends externally to the lower portion of the purgers funnel to a position near the lower surface of the housing. A venous tube is in fluid communication with the venous port and extends within the purgers funnel to a position near the lower surface of the housing.
- (16) In Example 11, the dual chamber blood reservoir of Example 10 in which the first vent tube extends downwardly within the upper portion of the purgers funnel and passes to an exterior of the purgers funnel through a first aperture formed in the central portion of the purgers funnel.
- (17) In Example 12, the dual chamber blood reservoir of Example 10 or 11 in which the first vent tube extends downwardly within the upper portion of the purgers funnel and passes to an exterior of the purgers funnel through a first aperture formed in the central portion of the purgers funnel.
- (18) In Example 13, the dual chamber blood reservoir of any of Examples 10 to 12, further including an elongate filter disposed within the housing such that the vent tubes and the venous tube extend downwardly through the elongate filter.
- (19) In Example 14, the dual chamber blood reservoir of Example 13 in which the elongate filter has a lower surface that is disposed near the lower surface of the housing.
- (20) In Example 15, the dual chamber blood reservoir of any of Examples 10 to 14, further including a plurality of purgers ports that pass through the cover and that are in fluid communication the upper portion of the purgers funnel.
- (21) Example 16 is a blood reservoir having a housing and a filtering assembly disposed within the housing. The housing has a top, a bottom, a venous inlet, a vent inlet and a purgers inlet. The filtering assembly extends from near the top of the housing to near the bottom of the housing. The

filtering assembly includes a support structure, a filter membrane disposed about the support structure and a defoamer that is disposed about the filter membrane. The filtering assembly includes a purgers funnel that is in fluid communication with the purgers inlet and that extends downwardly within the filter membrane. The filtering assembly includes a venous tube that is in fluid communication with the venous inlet and that extends through an interior of the purgers funnel to a location near a bottom surface of the filtering assembly. The filtering assembly also includes a vent tube that is in fluid communication with the vent inlet and that extends partially through an interior of the purgers funnel and partially exterior to the purgers funnel to a location near the bottom surface of the filtering assembly.

(22) In Example 17, the blood reservoir of Example 16 in which the venous tube and the vent tube extend downwardly within an interior space of the filter membrane.

(23) In Example 18, the blood reservoir of Example 16 or 17 in which the purgers inlet includes a plurality of purgers ports.

(24) Example 19 is an extracorporeal blood circuit that includes a heart lung machine, an oxygenator, a sampling line downstream of the oxygenator and a blood reservoir. The blood reservoir includes a vent blood inlet, a venous blood inlet and a purgers port configured to accept blood from the sampling line. The blood reservoir is configured to accommodate blood from the sampling line without causing excessive gaseous microembolic activity within the blood from the sampling line.

(25) In Example 20, the extracorporeal blood circuit of Example 19 in which the blood reservoir includes a purgers funnel that is in fluid communication with the purgers port, with the venous blood inlet extending downwardly through an interior of the purgers funnel such that blood from the sampling line is permitted to flow downwardly along an exterior surface of the venous blood inlet.

(26) While multiple embodiments are disclosed, still other embodiments of the present invention will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of an extracorporeal blood circuit in accordance with an embodiment of the present invention.

(2) FIG. 2 is a partially cross-sectioned perspective view of a blood reservoir in accordance with an embodiment of the present invention.

(3) FIG. 3A is a cross-sectional view of the blood reservoir of FIG. 2.

(4) FIG. 3B is a partially cross-sectioned perspective view of a blood reservoir in accordance with an embodiment of the present invention.

(5) FIG. 3C is a cross-sectional view of the blood reservoir of FIG. 3B.

(6) FIG. 4 is a perspective view of a purgers funnel in accordance with an embodiment of the present invention.

(7) FIG. 5 is a perspective view of a filtering assembly in accordance with an embodiment of the present invention.

(8) FIG. 6 is a cross-sectional view of the filtering assembly of FIG. 5.

(9) FIG. 7 is a perspective view of a portion of the filtering assembly of FIG. 5.

(10) FIG. 8 is a perspective view of a filtering assembly in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION

(11) FIG. 1 is a schematic illustration of an extracorporeal blood circuit **10**. As illustrated, the extracorporeal blood circuit **10** includes an HLM **12**, an oxygenator **14**, a sampling device **16** and a blood reservoir **18**. The HLM **12** is in fluid communication with a patient **20** and as such can receive blood from the patient **20** and moreover can return blood and other fluids to the patient **20**. The sampling device **16** may be a port or similar structure that permits blood to be withdrawn from the extracorporeal blood circuit **10** for lab work and/or additional testing done in the surgical arena. Blood in the sampling device **16** may flow into the blood reservoir **18** through a sampling line **22**.

(12) FIG. 2 is a partially cross-sectioned perspective view of a blood reservoir **24** that may be used as the blood reservoir **18** in the extracorporeal blood circuit **10** of FIG. 1. The blood reservoir **24** includes a clean (i.e., non-activated) section **26** and a dirty (i.e., activated) section **28**. In this, “clean” and “dirty” are relative terms pertaining to an expected level of solid particles or air bubbles within the blood entering each section. For example, vent blood and venous blood, which are usually fairly clean, may be processed within the non-activated section **26**, while suction blood, which tends to contain relatively more debris, may be processed within the activated section **28**.

(13) As shown in FIG. 2, the blood reservoir **24** includes a housing **30** and a cover **32**. A number of blood inlets, as will be described, extend through or are otherwise disposed within the cover **32**. The housing **30** includes a blood outlet **34** that may, in some embodiments, be in fluid communication with the HLM **12**. The housing **30** tapers to a bottom **46**. The cover **32** accommodates a venous inlet port **36**, one or more vent inlet ports **38** (only one is visible in this view) and a purgers inlet **40** having one or more purgers ports **42**. The cover **32** also accommodates a suction inlet **44**. In some embodiments, one or more of the venous inlet port **36**, the vent inlet port(s) **38**, the purgers inlet **40** or the suction inlet **44** may pass through the cover **32** such that they can rotate relative to the cover **32**.

(14) As shown, the non-activated section **26** includes a filtering assembly **48**, while the activated section **28** includes a filtering/defoaming assembly **50**. FIG. 3A is a cross-sectional view taken along line 3-3 of FIG. 2 and provides greater detail pertaining to the filtering assembly **48** and the filtering/defoaming assembly **50**. The blood reservoir **24** includes a movable or releasable valve **52** that, when in place as illustrated, keeps blood within the activated section **28** from entering the non-activated section **26**. In some cases, there may be a need for more blood than is available from the non-activated section **26** and thus the valve **52** may be lifted, rotated or otherwise moved to permit blood to pass from the activated section **28** to the non-activated section **26**.

(15) In some embodiments, the housing **30** may include a shield **54** that directs blood from the activated section **28** towards the bottom **46**. The shield **54** may be shaped and positioned to minimize turbulence within the blood flow. While relative blood levels may vary during use in the non-activated section **26** and the activated section **28** (when the valve **52** is closed), in some embodiments, the blood level within the non-activated section **26**, indicated by a line **56**, may be relatively lower than the blood level within the activated section **28**, as indicated by a line **58**. In some embodiments, the blood level within the non-activated section **26** may instead be higher than the blood level within the activated section **28**.

(16) In the activated section **28**, the suction filtering/defoaming assembly **50** includes several components. Blood from the suction inlet **44** may pass into a collection funnel **60** and may then slide or otherwise flow down a diverter **62** that is configured to minimize turbulence in the blood flow. The blood then passes through a cylindrical filter **64** and a defoamer **66** that is disposed about the cylindrical filter **64**. Blood thus filtered then collects within the activated section **28**, where it is stored until it is either needed or subsequently discarded through an exit port **68**.

(17) In the non-activated section **26**, the filtering assembly **48** includes several components, not all of which are visible in FIG. 3A. The filtering assembly **48** includes an elongate cylindrical filter **70** having a lower surface **72**. A venous inlet tube **74** that is in fluid communication with the venous inlet port **36** extends downwardly through an interior of the elongate cylindrical filter **70** and terminates at a position that is near the lower surface **72** of the elongate cylindrical filter **70**. A

cylindrical defoamer **76** is disposed about an upper region of the elongate cylindrical filter **70**.

(18) The filtering assembly **48** also includes a purgers funnel **78** that extends downwardly through the cylindrical defoamer **76** and into the elongate cylindrical filter **70**. The purgers funnel **78** is in fluid communication with the purgers inlet **40**. The venous inlet tube **74** extends downwardly through the purgers funnel **78**. In some embodiments, the venous inlet tube **74** has an outer diameter that is less than an inner diameter of the purgers funnel **78** such that purgers blood collected within the purgers funnel **78** may exit the purgers funnel **78** by sliding down an exterior of the venous inlet tube **74**. In some embodiments, this reduces turbulence in the flow of purgers blood, thereby reducing or even eliminating the formation of gaseous microembolic activity in the purgers blood. In some embodiments, the purgers funnel **78** may include fingers (not shown) that form an interference fit with the exterior of the venous inlet tube **74** yet permit blood to flow down the exterior of the venous inlet tube **74**. In some embodiments, any entrained air within the blood in the non-activated section **26** may travel up into the cylindrical defoamer **76**.

(19) FIG. 3B is a partially cross-sectioned perspective view blood reservoir **25** that may be used as the blood reservoir **18** in the extracorporeal blood circuit **10** of FIG. 1. In some embodiments, the blood reservoir **25** is similar in at least some constructional aspects to the blood reservoir **24**, and thus similar elements share reference numbers therebetween. The blood reservoir **25** includes a clean (i.e., non-activated) section **26** and a dirty (i.e., activated) section **28**. In this, “clean” and “dirty” are relative terms pertaining to an expected level of solid particles or air bubbles within the blood entering each section. For example, vent blood and venous blood, which are usually fairly clean, may be processed within the non-activated section **26**, while suction blood, which tends to contain relatively more debris, may be processed within the activated section **28**.

(20) As shown in FIG. 3B, the blood reservoir **25** includes a housing **30** and a cover **32**. A number of blood inlets, as will be described, extend through or are otherwise disposed within the cover **32**. The housing **30** includes a blood outlet **34** that may, in some embodiments, be in fluid communication with the HLM **12**. The housing **30** tapers to a bottom **46**. The cover **32** accommodates a venous inlet port **36**, one or more vent inlet ports **38** (only one is visible in this view) and a purgers inlet **40** having one or more purgers ports **42**. The cover **32** also accommodates a suction inlet **44**. In some embodiments, one or more of the venous inlet port **36**, the vent inlet port(s) **38**, the purgers inlet **40** or the suction inlet **44** may pass through the cover **32** such that they can rotate relative to the cover **32**. As shown, the non-activated section **26** includes a filtering assembly **48**, while the activated section **28** includes a filtering/defoaming assembly **50**.

(21) FIG. 3C is a cross-sectional view taken along line 3'-3' of FIG. 3B and provides greater detail pertaining to the filtering assembly **48** and the filtering/defoaming assembly **50**. The blood reservoir **25** includes a movable or releasable valve **52** that, when in place as illustrated, keeps blood within the activated section **28** from entering the non-activated section **26**. In some cases, there may be a need for more blood than is available from the non-activated section **26** and thus the valve **52** may be lifted, rotated or otherwise moved to permit blood to pass from the activated section **28** to the non-activated section **26**.

(22) In some embodiments, the housing **30** may include a shield **55** that directs blood from the activated section **28** towards the bottom **46**. The shield **55** may be shaped and positioned to minimize turbulence within the blood flow. In some embodiments, as illustrated, the shield **55** may include a frame portion **57** and a porous media portion **59**. The frame portion **57** supports the porous media portion **59** and helps to anchor the shield **55** within the housing **30**. The porous media portion **59** slows blood passing through the shield **55**.

(23) While relative blood levels may vary during use in the non-activated section **26** and the activated section **28** (when the barrier **52** is closed), in some embodiments, the blood level within the non-activated section **26**, indicated by a line **56**, may be relatively lower than the blood level within the activated section **28**, as indicated by a line **58**. In some embodiments, the blood level within the non-activated section **26** may instead be higher than the blood level within the activated

section **28**.

(24) In the activated section **28**, the suction filtering/defoaming assembly **50** includes several components. Blood from the suction inlet **44** may pass into a collection funnel **60** and may then slide or otherwise flow down a diverter **62** that is configured to minimize turbulence in the blood flow. The blood then passes through a cylindrical filter **64** and a defoamer **66** that is disposed about the cylindrical filter **64**. Blood thus filtered then collects within the activated section **28**, where it is stored until it is either needed or subsequently discarded through an exit port **68**. In some embodiments, blood stored within the activated section **28** may be released into the non-activated section **26** by opening the valve **52**.

(25) In the non-activated section **26**, the filtering assembly **48** includes several components, not all of which are visible in FIG. 3A. The filtering assembly **48** includes an elongate cylindrical filter **70** having a lower surface **72**. A venous inlet tube **74** that is in fluid communication with the venous inlet port **36** extends downwardly through an interior of the elongate cylindrical filter **70** and terminates at a position that is near the lower surface **72** of the elongate cylindrical filter **70**. A cylindrical defoamer **76** is disposed about an upper region of the elongate cylindrical filter **70**.

(26) The filtering assembly **48** also includes a purgers funnel **78** that extends downwardly through the cylindrical defoamer **76** and into the elongate cylindrical filter **70**. The purgers funnel **78** is in fluid communication with the purgers inlet **40**. The venous inlet tube **74** extends downwardly through the purgers funnel **78**. In some embodiments, the venous inlet tube **74** has an outer diameter that is less than an inner diameter of the purgers funnel **78** such that purgers blood collected within the purgers funnel **78** may exit the purgers funnel **78** by sliding down an exterior of the venous inlet tube **74**. In some embodiments, this reduces turbulence in the flow of purgers blood, thereby reducing or even eliminating the formation of gaseous microembolic activity in the purgers blood. In some embodiments, the purgers funnel **78** may include fingers (not shown) that form an interference fit with the exterior of the venous inlet tube **74** yet permit blood to flow down the exterior of the venous inlet tube **74**. In some embodiments, any entrained air within the blood in the non-activated section **26** may travel up into the cylindrical defoamer **76**.

(27) FIG. 4 is a perspective view of an embodiment of the purgers funnel **78**. In the illustrated embodiment, the purgers funnel **78** includes an upper portion **80**, a lower portion **82** and a tapered central portion **84** between the upper portion **80** and the lower portion **82**. In some embodiments, the upper portion **80** may be conical or otherwise tapered in shape. In some cases, the lower portion **82** may be cylindrical in shape. In the illustrated embodiment, the central portion **84** of the purgers funnel **78** includes a first aperture **86** and a second aperture **88**. The first aperture **86** and the second aperture **88** may be configured to permit first and second vent tubes (illustrated in a subsequent Figure) to pass therethrough. In some embodiments, the first aperture **86** and the second aperture **88** may be radially spaced about 180 degrees apart.

(28) FIG. 5 is a perspective view of the filtering assembly **48**. The filtering assembly **48** includes, as shown in FIG. 3A, the elongate cylindrical filter **70** and the cylindrical defoamer **76**. The elongate cylindrical filter **70** includes a filter membrane **90** and a support structure **92**. As illustrated, the filter membrane **90** is disposed inside of the support structure **92**. In some embodiments, the filter membrane **90** may instead be disposed about the support structure **92**. The support structure **92** may provide sufficient support to the filter membrane **90** to hold the filter membrane **90** in a desired configuration against the fluid pressures to which the filter membrane **90** may be exposed during operation of the blood reservoir **24**.

(29) FIG. 6 is a cross-sectional view taken along line 6-6 of FIG. 5 and illustrates a blood flow path for purgers blood. As indicated by arrows **94**, purge blood may enter the blood reservoir **24** through the purgers ports **42**. The purgers blood then travels down through the purgers funnel **78** as indicated by arrows **96**, and exits through a bottom **98** of the purgers funnel **78**. As indicated by arrows **100**, the blood then slides or otherwise flows down the exterior surface of the venous inlet tube **74**.

(30) FIG. 7 is a perspective view of a portion of the filtering assembly 48, illustrating the venous inlet tube 74, a first vent tube 102 and a second vent tube 104. The venous inlet tube 74, the first vent tube 102 and the second vent tube 104 extend downwardly from the cover 32 through an interior of the elongate cylindrical filter 70. As shown in FIG. 4, the first vent tube 102 may pass through the first aperture 86 and the second vent tube 104 may pass through the second aperture 88. The first vent tube 102 may be considered as extending within the purgers funnel 78 above the first aperture 86 but exterior to the purgers funnel 78 below the first aperture 86. Similarly, the second vent tube 104 may be considered as extending within the purgers funnel 78 above the second aperture 88 but exterior to the purgers funnel 78 below the second aperture 88. In some embodiments, the venous inlet tube 74, the first vent tube 102 and the second vent tube 104 each extend downwardly to a position that is proximate or near to the lower surface 72 of the elongate cylindrical filter 70. As a result, in some embodiments, turbulence and resulting blood cell damage may be reduced or eliminated.

(31) FIG. 8 is a perspective view of an embodiment of the filtering/defoaming assembly 50. In some embodiments, the filtering/defoaming assembly 50 includes a plastic frame 150 that supports the filtering/defoaming assembly 50 and provides the filtering/defoaming assembly 50 with an annular or ovoid shape. A foam cylinder such as a polyurethane foam cylinder 152 is disposed within the plastic frame 150 and at least partially defines an internal sliding surface 154. An outer surface of the foam cylinder 152 is at least partially wrapped in a polyester felt 156. In some embodiments, the polyester felt 156 has a pore size of about 40 microns.

(32) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. For example, while the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the above described features.

Claims

1. A blood reservoir for use with a heart lung machine, the blood reservoir comprising: a housing including a venous blood inlet, a purgers port for accepting blood from a sampling line, a purgers funnel in fluid communication with the purgers port, and a defoamer disposed such that the purgers funnel extends through the defoamer and at least part of the purgers funnel extends below the defoamer, wherein the venous blood inlet extends downwardly through an interior of the purgers funnel, the purgers funnel being configured to permit the blood from the sampling line to exit the purgers funnel and flow downwardly along an exterior surface of the venous blood inlet.
2. The blood reservoir of claim 1, wherein the purgers funnel has an upper portion, a lower portion and a central portion, wherein the upper portion of the purgers funnel is in fluid communication with the purgers port.
3. The blood reservoir of claim 2, further comprising: a venous tube in fluid communication with the venous blood inlet and extending within the purgers funnel, such that the lower portion of the purgers funnel is configured to extend alongside a portion of the venous tube.
4. The blood reservoir of claim 3, wherein the lower portion of the purgers funnel has an inner diameter that is greater than an outer diameter of the venous tube such that blood exiting the purgers funnel slides down an exterior surface of the venous tube.
5. The blood reservoir of claim 4, further comprising: an elongate filter disposed such that the venous tube extends downwardly inside the elongate filter.
6. The blood reservoir of claim 5, wherein the elongate filter extends to near a lower surface of the housing.
7. The blood reservoir of claim 6, wherein the elongate filter extends beyond a lower end of the venous tube.

8. The blood reservoir of claim 2, further comprising: a first vent blood inlet, and a first vent tube in fluid communication with the first vent blood inlet and extending external to the lower portion of the purgers funnel, wherein the first vent tube extends downwardly within the upper portion of the purgers funnel and passes to an exterior thereof through a first aperture formed in the central portion of the purgers funnel.
 9. The blood reservoir of claim 8, further comprising: a second vent blood inlet, and a second vent tube in fluid communication with the second vent blood inlet and extending external to the lower portion of the purgers funnel, wherein the second vent tube extends downwardly within the upper portion of the purgers funnel and passes to an exterior thereof through a second aperture formed in the central portion of the purgers funnel.
 10. A blood reservoir for use with a heart lung machine, the blood reservoir comprising: a housing including a purgers port for accepting blood from a sampling line, a purgers funnel, and a defoamer, wherein the purgers funnel includes an upper portion, a lower portion and a central portion, the upper portion in fluid communication with the purgers port, wherein the defoamer is positioned so the purgers funnel extends through the defoamer and at least part of the lower portion of the purgers funnel extends below the defoamer.
 11. The blood reservoir of claim 10, further comprising: a venous blood inlet tube extending downwardly through an interior of the purgers funnel, wherein the purgers funnel is configured to permit the blood from the sampling line to flow downwardly along an exterior surface of the venous blood inlet tube.
 12. The blood reservoir of claim 11, wherein the venous blood inlet tube extends downwardly within the upper portion of the purgers funnel and passes to an exterior thereof through a first aperture formed in the central portion of the purgers funnel.
 13. The blood reservoir of claim 12, further comprising: an elongate filter disposed around at least the lower portion of the purgers funnel.
 14. The blood reservoir of claim 13, wherein the elongate filter extends to near a lower surface of the housing.
 15. The blood reservoir of claim 14, wherein the elongate filter extends beyond a lower end of the venous blood inlet tube.
 16. The blood reservoir of claim 10, further comprising: a shield, wherein the blood reservoir includes an activated section and a non-activated section, and the shield is situated between the activated section and the non-activated section and configured to direct blood from the activated section to the non-activated section.
 17. The blood reservoir of claim 16, wherein the shield includes a frame and a porous media disposed within the frame to dissipate velocity of blood flowing from the activated section to the non-activated section.
 18. The blood reservoir of claim 16, further comprising: a releasable valve configured to keep blood within the activated section from entering the non-activated section until the valve is released.
 19. The blood reservoir of claim 16, wherein the activated section includes a filtering and defoaming assembly.
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